



Department of Applied Computing and Artificial Intelligence

Faculty of Computing

SECR1213 NETWORK COMMUNICATIONS

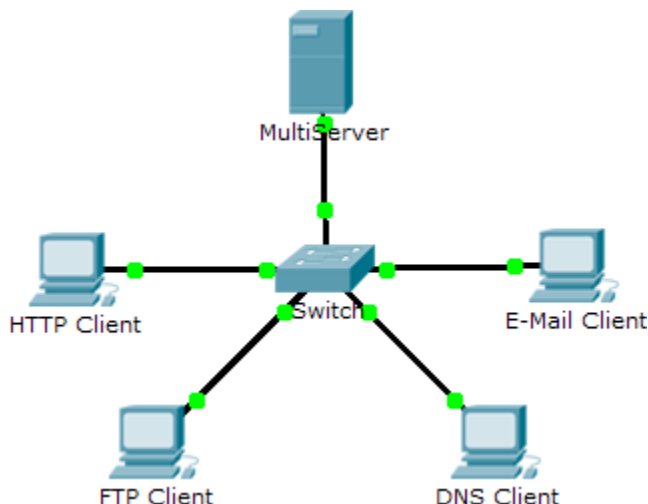
Lab 2

Packet Tracer Simulation – TCP and UDP communications (4%)

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Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.255

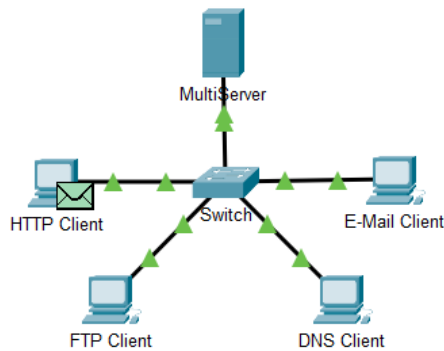
Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 4, Received = 16, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Step 2: Generate web (HTTP) traffic.

- Switch to Simulation mode.
- Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- Minimize, but do not close, the **HTTP Client** configuration window.



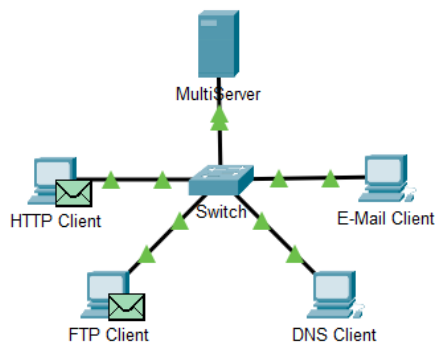
Event List		
Vis.	Time(sec)	Last Device
	0.000	--

Step 3: Generate FTP traffic.

- Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- Minimize, but do not close, the **FTP Client** configuration window.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.254
Trying to connect...192.168.1.254
Connected to 192.168.1.254
220- Welcome to PT Ftp server
Username:|
  
```

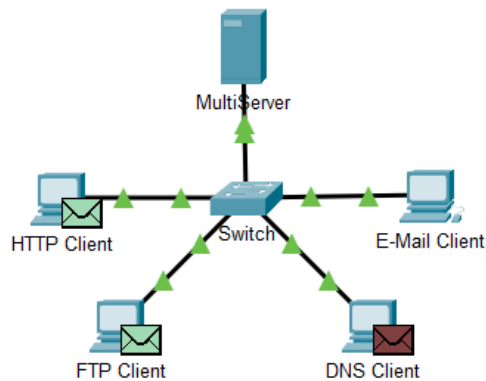


Simulation Panel		
Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--

Step 4: Generate DNS traffic.

- Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- Minimize, but do not close, the **DNS Client** configuration window.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>nslookup multiserver.pt.ptu
```



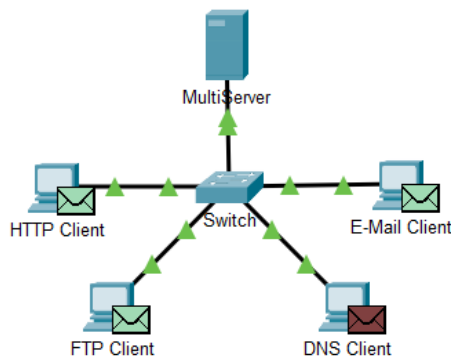
Simulation Panel		
Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--

Step 5: Generate Email traffic.

- Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- Click **Compose** and enter the following information:
 - To:** user@multiserver.pt.ptu
 - Subject:** Personalize the subject line
 - E-Mail Body:** Personalize the Email
- Click **Send**.
- Minimize, but do not close, the **E-Mail Client** configuration window.

Sending mail to user@multiserver.pt.ptu , with subject : Personalize the subject line .. Mail Server: 192.168.1.254

Cancel
Send/Receive



Simulation Panel		
Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

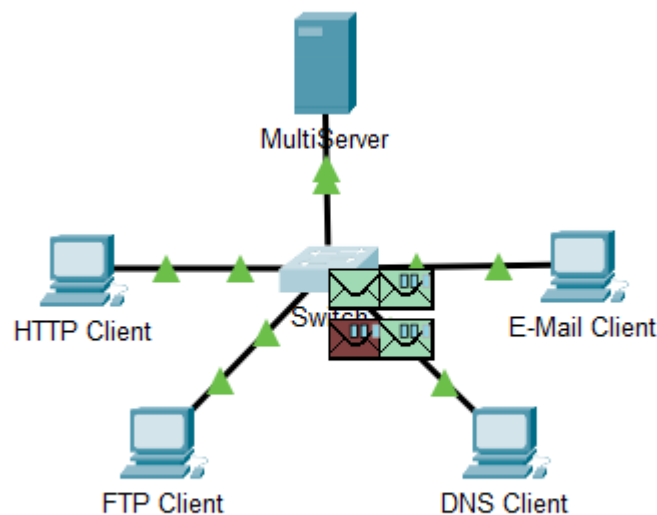
Simulation Panel		
Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--

Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

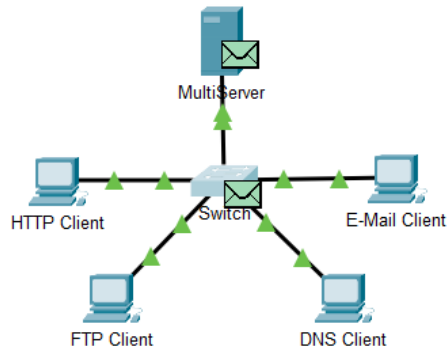
You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- Click **Capture/Forward** once. All of the PDUs are transferred to the switch.



- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

The PDUs are stored in switch.

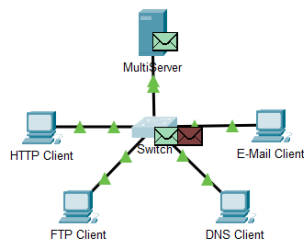


Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--
	0.001	HTTP Client
	0.001	FTP Client
	0.001	DNS Client
	0.001	E-Mail Client
	0.002	Switch
	0.002	--

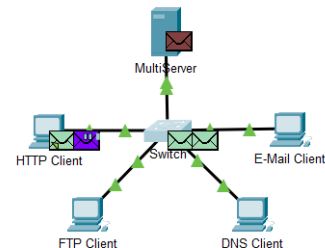
- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

Multiplexing

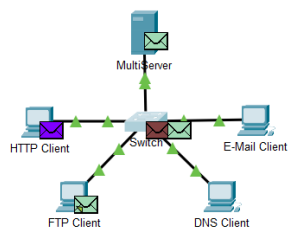
1.



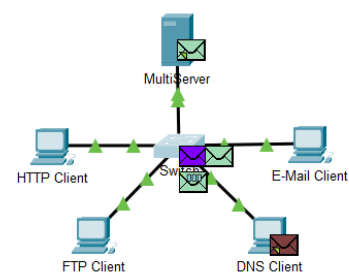
2.



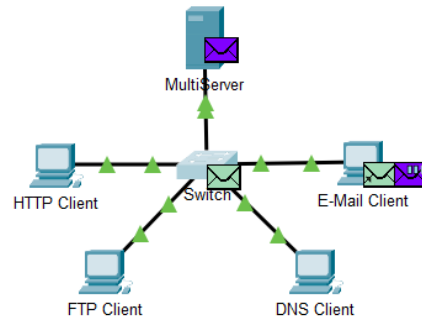
3.



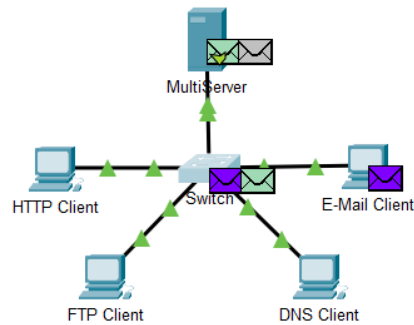
4.



5.



6.



- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?

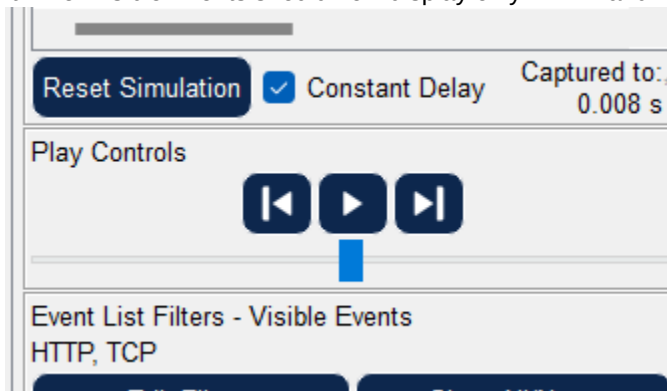
Because of different protocol

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

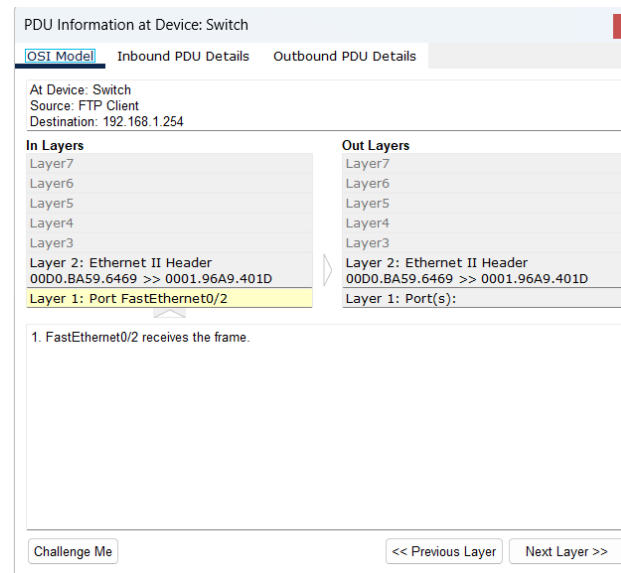
Step 2: Examine HTTP traffic as the clients communicate with the server.

- Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
 - Click **Edit Filters** and toggle the **Show All/None** check box.
 - Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.



- Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP



TCP		E	
0	4	8	16
SOURCE PORT:1025		DESTINATION PORT:21	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OFFSET :0x0	RESER VED: 0	FLAGS:0b0000001 0	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

Are these communications considered to be reliable?

Yes. TCP is a reliable transport which has a connection between the sending and receiving process.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1025, DEST PORT: 80, SEQUENCE NUM: 0, ACK NUM:0

FLAGS=0B000010=SYN

SYN is written in the field to the left of the **WINDOW** field.

TCP		Bits	
0	4	8	16
SOURCE PORT:1025		DESTINATION PORT:80	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OFFSET:0x0	RESERVED:0	FLAGS:0b00000001 0	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

- Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.
- Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 80, DEST PORT: 1025, SEQUENCE NUM: 0, ACK NUM:1

FLAGS=0b00010010=SYN+ACK

SRC PORT and DEST PORT are reversed, ACK NUM change to 1 and SYN changed to SYN+ACK.

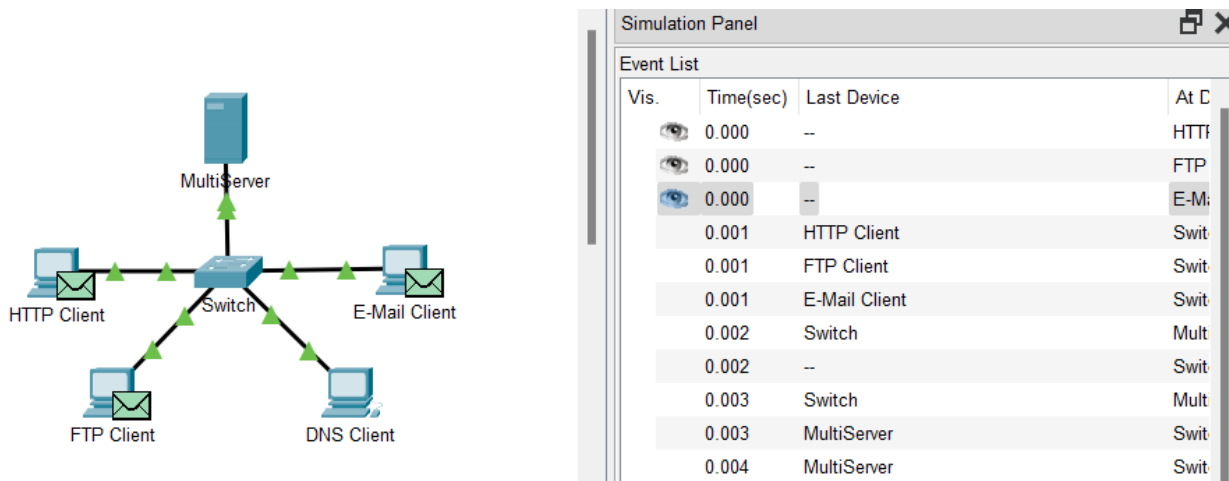
TCP																								Bits							
0		4		8		16		24																							
SOURCE PORT:80												DESTINATION PORT:1025																			
SEQUENCE NUMBER:0																															
ACKNOWLEDGEMENT NUMBER:1																															
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010010								WINDOW:16384															
CHECKSUM:0x0000												URGENT POINTER:0x0000																			
OPTION																															
DATA (VARIABLE LENGTH)																		PADDING: 0													

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

SRC PORT: 1025, DEST PORT: 80, SEQUENCE NUM: 1, ACK NUM:1
Value of SRC PORT and DEST PORT are reversed, and the SEQUENCE NUM and ACK NUM are 1.
SYN and SYN+ACK changed to PSH+ACK.

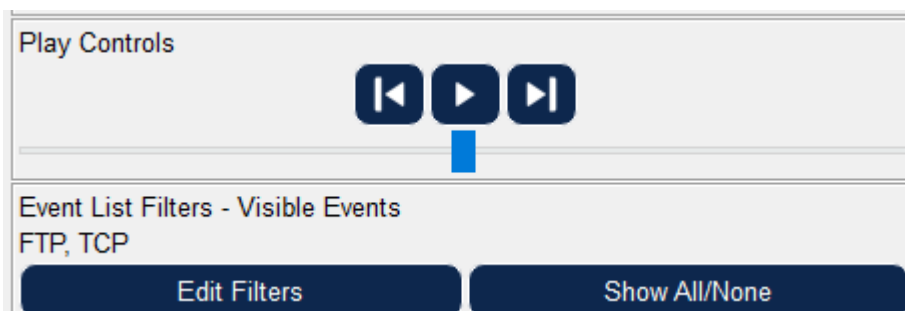
TCP																												Bits			
0		4		8								16								24											
SOURCE PORT:1025														DESTINATION PORT:80																	
SEQUENCE NUMBER:1																															
ACKNOWLEDGEMENT NUMBER:1																															
OFFSET :0x0				RESER VED: 0				FLAGS:0b0001100 0								WINDOW:65535															
CHECKSUM:0x0000														URGENT POINTER:0x0000																	
OPTION																															
DATA (VARIABLE LENGTH)																						PADDING: 0									

- i. Click **Back** until the simulation is reset.



Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.



- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

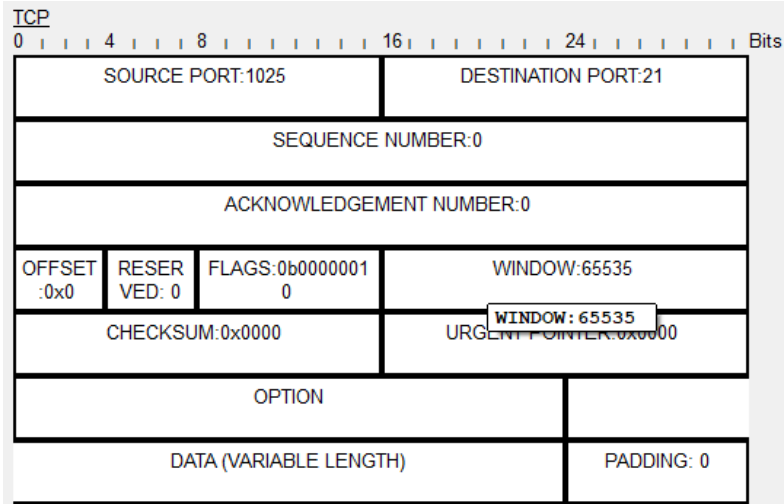
TCP

Are these communications considered to be reliable?

Yes, because it establishes a connection before data transfer and ensure data arrives in order

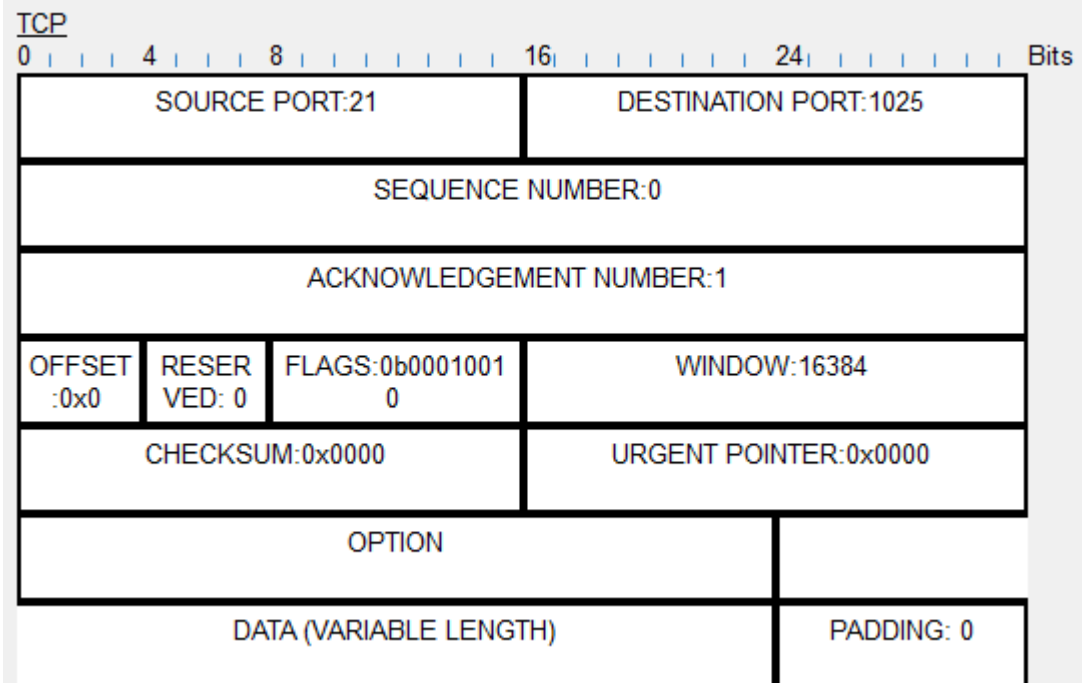
- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1025, DEST PORT: 21, SEQUENCE NUM: 0, ACK NUM:0
SYN is written in the field to the left of the WINDOW field.



- Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.
- Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 21, DEST PORT: 1025, SEQUENCE NUM: 0, ACK NUM:1
SRC PORT and DEST PORT are reversed, and the ACK NUM becomes 1.
SYN changed to SYN+ACK.



- Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

SRC PORT: 1025, DEST PORT: 21, SEQUENCE NUM: 1, ACK NUM:1
SRC PORT and DEST PORT are reversed, and the SEQUENCE NUM and ACK NUM are 1.
SYN and SYN+ACK changed to ACK.

TCP																							
0				4				8				16				24				Bits			
SOURCE PORT:1025												DESTINATION PORT:21											
SEQUENCE NUMBER:1																							
ACKNOWLEDGEMENT NUMBER:1																							
OFFSET:0x0				RESERVED: 0				FLAGS:0b00010000 0								WINDOW:65535							
CHECKSUM:0x0000												URGENT POINTER:0x0000											
OPTION																							
DATA (VARIABLE LENGTH)																		PADDING: 0					

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.
- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

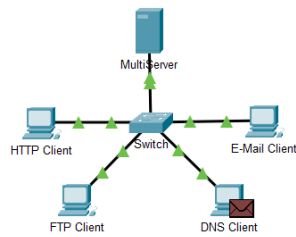
Welcome to PT Ftp server

FTP Response																			
0				4															Bytes
Code:220																			
Message:Welcome to PT Ftp server																			

- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.



Simulation Panel			
Event List			
Vis.	Time(sec)	Last Device	At Device
☒	0.000	--	DNS Client
	0.001	DNS Client	Switch
	0.003	--	Switch
	0.004	Switch	MultiServer
	0.005	MultiServer	Switch
	0.006	Switch	DNS Client

☒ Constant Delay
 Captured to: 0.008 s

Play Controls

Event List Filters - Visible Events
DNS, UDP

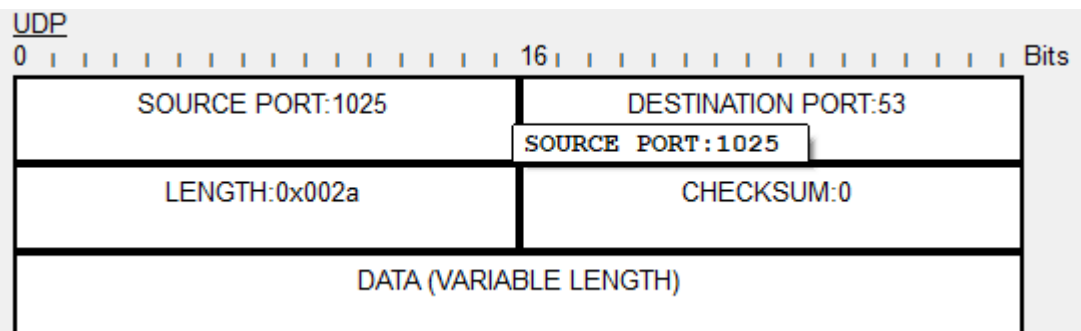
- Click the PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

UDP

Are these communications considered to be reliable? **No because it sends data without setting up a connection or ensuring delivery.**

- Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

SRC PORT: 1025, DEST PORT: 53. There is no SEQUENCE and ACK NUM. This is because UDP no need to build a reliable connection.

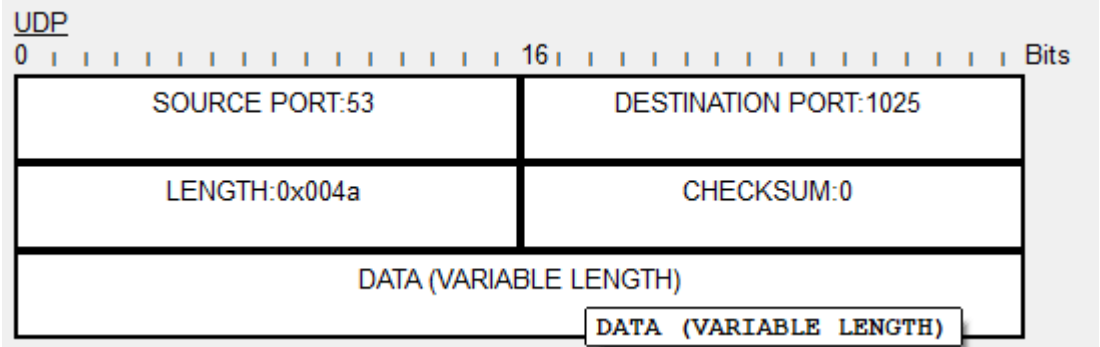


- Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

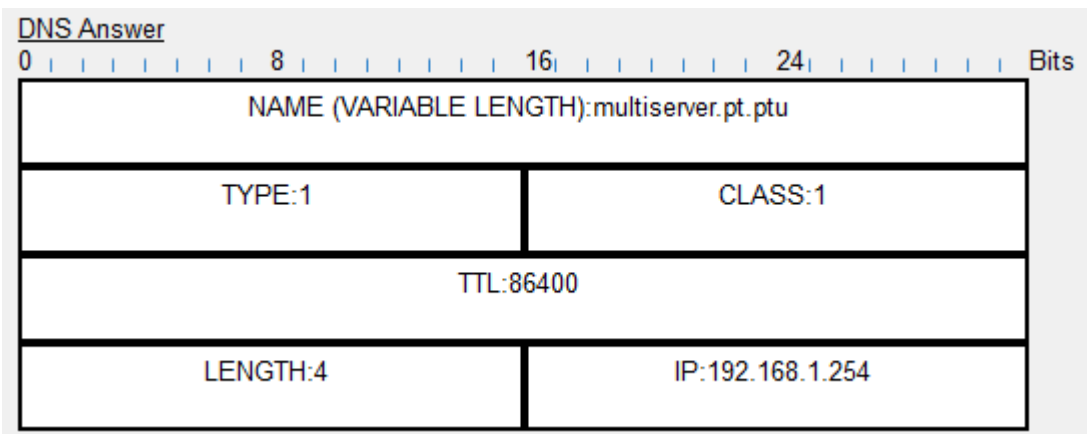
SRC PORT: 53, DEST PORT: 1025.

SRC PORT and DEST PORT are reversed.

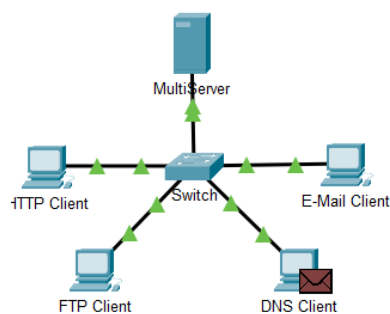


- g. What is the last section of the **PDU** called?

DNS Answer



- h. Click **Back** until the simulation is reset.



Step 5: Examine email traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

TCP

Are these communications considered to be reliable?

YES, because it establishes a connection before data transfer and ensures data arrives in

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

SRC PORT: 1025, DEST PORT: 25, SEQUENCE NUM: 0, ACK NUM:0
SYN is written in the field to the left of the **WINDOW** field.

TCP		Bits	
0	4	8	16
SOURCE PORT:1025		DESTINATION PORT:25	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OFFSET:0x0	RESERVED: 0	FLAGS:0b00000010	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.
- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

SRC PORT: 25, DEST PORT: 1025, SEQUENCE NUM: 0, ACK NUM:1

FLAGS=0B00010010=SYN+ACK

SRC PORT and DEST PORT are reversed, and the ACK NUM becomes 1. FLAGS changed to SYN+ACK.

TCP												Bits
0		4		8		16		24				
SOURCE PORT:25						DESTINATION PORT:1025						
SEQUENCE NUMBER:0												
ACKNOWLEDGEMENT NUMBER:1												
OFFSET:0x0		RESERVED: 0		FLAGS:0b000100100				WINDOW:16384				
CHECKSUM:0x0000						URGENT POINTER:0x0000						
OPTION												
DATA (VARIABLE LENGTH)										PADDING: 0		

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

SRC PORT: 1025, DEST PORT: 25, SEQUENCE NUM: 1, ACK NUM:1

SRC PORT and DEST PORT are reversed, and the SEQUENCE NUM and ACK NUM are 1. SYN and SYN+ACK changed to ACK.

TCP												Bits
0	4	8	16	24								
SOURCE PORT:1025						DESTINATION PORT:25						
SEQUENCE NUMBER:1												
ACKNOWLEDGEMENT NUMBER:1												
OFFSET :0x0		RESER VED: 0		FLAGS:0b0001000 0				WINDOW:65535				
CHECKSUM:0x0000						URGENT POINTER:0x0000						
OPTION												
DATA (VARIABLE LENGTH)									PADDING: 0			

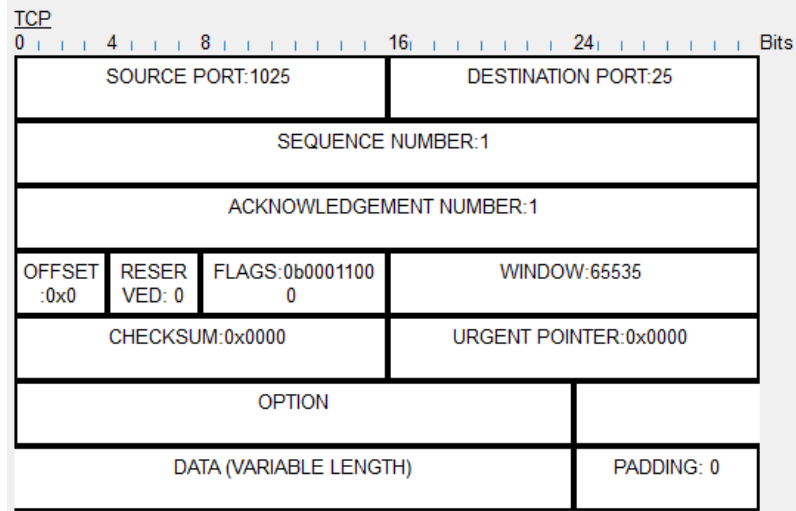
- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

- i. How are the port and sequence numbers different from the previous two **PDUs**?

SRC PORT: 1025, DEST PORT: 25, SEQUENCE NUM: 1, ACK NUM:1

FLAGS=0b00011000=PSH+ACK

SRC PORT and DEST PORT are reversed, and the SEQUENCE NUM and ACK NUM are 1. FLAGS changed to PSH+ACK.



- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
STMP is associated with TCP port 25, and POP3 is associated with TCP port 110.
- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:
 - 1) Switch back to **Realtime** mode.
 - 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **netstat** command. What protocols are listed in the left column?

TCP

```
C:\>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
```

What port numbers are being used by the server?

port 21

```
C:\>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
```

- c. What states are the sessions in?

ESTABLISHED

```
C:\>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP   192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
```

- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

FTP Client(PORT 21)

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

The server is waiting for username input from the client.

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.1.254
Trying to connect...192.168.1.254
Connected to 192.168.1.254
220- Welcome to PT Ftp server
Username:
```

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	